By the way...

I carried out 4,700 experiments with different materials to find the perfect filament, including hair from a beard.

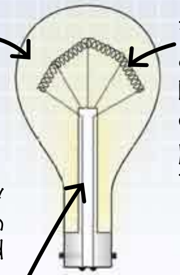
Edison's light-bulb moment

Meanwhile, in the United States, ***Thomas Edison*** was experimenting with light bulbs too. He realized that the **FILAMENT**—the part that gets hot and glows—was the key to long-lasting light. By 1880, his charred bamboo filaments were burning for more than 1,200 hours. After falling out over who invented what, **Edison and Swan joined forces**. Soon, they were bringing light to everybody.

Thomas Edison said, "We will make electricty so cheap that only the rich will burn candles."

HOW IT WORKS

A light bulb's filament is made out of material that does not conduct electricity very well. This resistance to the current makes the filament heat up and radiate light. The bulb is filled with nonreactive gases so that the hot filament does not catch fire.



Plugging in

POWER PLANTS were set up so electricity could reach everybody with Edison's bulbs. The world's first ***electric company*** started out with 52 customers in 1882. Before long, people were finding their way home in the dark by the glow of **electric streetlights**, and flipping switches for lights in their homes. This advertisement for Edison's bulbs dates from 1909.

How it changed the world

Light bulbs meant safe, bright lighting at the flip of a switch. Only now, after more than a hundred years, is the basic design being improved to make it more efficient.

FLASHBULBS *were invented in 1929, replacing dangerous and noisy* **FLASH POWDER**.

The first practical **LIGHT-EMITTING DIODE** *(LED) was developed in 1962. They can be used as a replacement for light bulbs.*